

Math 230B Lecture Notes

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1 Week 5

Example ($SL(n, \mathbb{F})$). Suppose $\mathbb{F} = \mathbb{R}$ (or $\mathbb{C}$). We claim that

$$SL(n, \mathbb{F}) = \{A \in GL(n, \mathbb{F}) : \det A = 1\}$$

is a Lie subgroup of $GL(n, \mathbb{F})$.

Indeed, we have that

- $SL(n, \mathbb{F})$ is closed under multiplication, indeed, if $A, B \in SL(n, \mathbb{F})$, then

$$\det(AB) = (\det A)(\det B) = 1 \cdot 1 = 1.$$

So, AB is invertible and $\det(AB) = 1$. This implies that

$$AB \in SL(n, \mathbb{F}).$$

- $SL(n, \mathbb{F})$ contains the inverse of each of its elements. Indeed, suppose $A \in SL(n, \mathbb{F})$, then A is invertible and

$$\det(A^{-1}) = \frac{1}{\det A} = \frac{1}{1} = 1.$$

So, $A^{-1} \in SL(n, \mathbb{F})$.

From the above, we can conclude that $SL(n, \mathbb{F})$ is a subgroup of $GL(n, \mathbb{F})$.

2 Week 6

Example (Special Orthogonal Group). Note that

$$\begin{aligned} SO(n) &= \{A \in M(n, \mathbb{R}) : A^T A = I \text{ and } \det A > 0\} \\ &= \{A \in M(n, \mathbb{R}) : A^T A = I \text{ and } \det A = 1\} \end{aligned}$$

We will prove that $SO(n)$ is a subgroup of $O(n)$.

- Note that $SO(n)$ is closed under multiplication. Indeed, suppose A and B are in $SO(n)$. Our goal is to show that $AB \in SO(n)$. Then we have

$$A, B \in SO(n) \implies A, B \in O(n) \implies AB \in O(n).$$

Since $A, B \in SO(n)$, it follows that $\det A = 1$ and $\det B = 1$. Hence,

$$\det AB = (\det A)(\det B) = 1 \cdot 1 = 1.$$

- Also observe that $SO(n)$ contains the inverse of each of its elements. Indeed, if $A \in SO(n)$, then

$$A \in SO(n) \implies A \in O(n) \implies A^{-1} \in O(n). \quad (1)$$

Furthermore,

$$A \in SO(n) \implies \det A = 1 \implies \det A^{-1} = \frac{1}{1} = 1 \implies A^{-1} \in SO(n) \quad (2)$$

From (1) and (2), we can see that $SO(n)$ is a subgroup of $O(n)$. Now, we need to find out whether $SO(n)$ is a Lie subgroup of $O(n)$. Note that $SO(n) = O(n) \cap \det^{-1}(0, \infty)$ where $(0, \infty)$ is an open set, $\det$ is a continuous function, and the preimage of an open set will be open in $M(n, \mathbb{R})$. Hence, $SO(n)$ will be open in $O(n)$. Thus, $SO(n)$ is an embedded submanifold of $O(n)$ with dimension $\frac{n^2-n}{2}$. Thus, $SO(n)$ is a Lie subgroup of $O(n)$ with dimension $\frac{n^2-n}{2}$.

Example (Special Unitary Group). Recall that

$$SU(n) = \{A \in M(n, \mathbb{C}) : A^* A = I \text{ and } \det A = 1\}.$$

We will show that the above is also a subgroup of $U(n)$. Clearly, we can see that $SU(n) \subseteq U(n)$.

- $SU(n)$ is closed under multiplication. Indeed, suppose A and B are in $SU(n)$. Then observe

that

$$A, B \in \text{SU}(n) \implies A, B \in \text{U}(n) \implies AB \in \text{U}(n) \quad (1)$$

and

$$\begin{aligned} A, B \in \text{SU}(n) &\implies \det A = 1 \text{ and } \det B = 1 \\ &\implies \det AB = 1 \\ &\implies AB \in \text{SU}(n). \end{aligned}$$

- Also, $\text{SU}(n)$ contains the inverse of each of its elements. Suppose $A \in \text{SU}(n)$. Then

$$A \in \text{SU}(n) \implies A \in \text{U}(n) \implies A^{-1} \in \text{U}(n) \quad (3)$$

and

$$A \in \text{SU}(n) \implies \det A = 1 \implies \det A^{-1} = \frac{1}{\det A} = \frac{1}{1} = 1 \quad (4)$$

Now, (3) and (4) imply that $A^{-1} \in \text{SU}(n)$.

Now, we need to find out if $\text{SU}(n)$ a Lie subgroup of $\text{U}(n)$? Of course! Let $F : \text{U}(n) \rightarrow \mathbb{R}$ be defined as follows:

$$F(A) = \varphi.$$

where we previously proved that if $A \in \text{U}(n)$, then $|\det A| = 1$. So, there exists an angle $\varphi \in (-\pi, \pi]$ such that $\det A = e^{i\varphi}$. It can be shown that

$$\forall A \in \text{U}(n) \quad \text{rank} DF(A) = 1.$$

Thus, $\text{SU}(n) = F^{-1}(0)$ is an $n^2 - 1$ dimensional embedded submanifold of $\text{U}(n)$. Therefore, $\text{SU}(n)$ is a Lie subgroup of $\text{U}(n)$ with dimension $n^2 - 1$.

Definition (Bilinear Form). A function $B : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ is said to be a **bilinear form** on $\mathbb{R}^n$ if it satisfies the following condition:

- (1) $\forall x, y, z \in \mathbb{R}^n \quad B(x + y, z) = B(x, z) + B(y, z)$
- (2) $\forall c \in \mathbb{R} \text{ and } \forall x, y \in \mathbb{R}^n \quad B(cx, y) = cB(x, y)$
- (3) $\forall x, y, z \in \mathbb{R}^n \quad B(x, y + z) = B(x, y) + B(x, z)$
- (4) $\forall c \in \mathbb{R} \forall x, y \in \mathbb{R}^n \quad B(x, cy) = cb(x, y)$

Moreover,

- The bilinear form B is said to be **symmetric** if

$$\forall x, y \in \mathbb{R}^n \quad B(x, y) = B(y, x).$$

- The bilinear form B is said to be **skew-symmetric** if

$$\forall x, y \in \mathbb{R}^n \quad B(x, y) = -B(y, x).$$

- The bilinear form B is said to be **non-degenerate** if the following statement is true

$$B(x, y) = 0 \quad \forall y \in \mathbb{R}^n \implies x = 0.$$

- A symmetric bilinear form B is said to be **positive definite** if

$$B(x, x) > 0 \quad \forall x \neq 0.$$

Example (Standard Inner Product on $\mathbb{R}^n$). The standard inner product in $\mathbb{R}^n$

$$\langle x, y \rangle = \sum_{i=1}^n x_i y_i$$

is a symmetric positive definite bilinear form on $\mathbb{R}^n$.

For example, let's verify condition (1) for bilinearity:

$$\begin{aligned} \langle x + y, z \rangle &= \sum_{i=1}^n [(x_i + y_i) z_i] \\ &= \sum_{i=1}^n [x_i z_i + y_i z_i] \\ &= \sum_{i=1}^n x_i z_i + \sum_{i=1}^n y_i z_i \\ &= \langle x, z \rangle + \langle y, z \rangle. \end{aligned}$$

I leave you to check the other conditions yourself.

In general, a symmetric positive definite bilinear form on a real vector space V , by definition, is called an **inner product** on V .

Example (Standard Symplectic Form on $\mathbb{R}^{2n}$). Define $w : \mathbb{R}^{2n} \times \mathbb{R}^{2n} \rightarrow \mathbb{R}$ as follows:

$$\forall x, y \in \mathbb{R}^{2n} \quad w(x, y) = \sum_{j=1}^n x_j y_{n+j} - x_{n+j} y_j.$$

For example,

$$\begin{aligned} n = 1 &\implies w(x, y) = x_1 y_2 + x_2 y_1 \\ n = 2 &\implies w(x, y) = (x_1 y_3 - x_3 y_1) + (x_2 y_4 - x_4 y_2) \\ &\vdots \end{aligned}$$

If we let $\Omega = \begin{pmatrix} O_{n \times n} & I_{n \times n} \\ -I_{n \times n} & O_{n \times n} \end{pmatrix}$, then we can write the standard symplectic form in the following way:

$$w(x, y) = \langle x, \Omega y \rangle$$

where $\langle, \rangle$ is the standard inner product in $\mathbb{R}^{2n}$.

Proposition (Preserving the standard symplectic form on $\mathbb{R}^{2n}$). Suppose $A \in M(2n, \mathbb{R})$. We say that A preserves the standard symplectic form on $\mathbb{R}^{2n}$ if and only if $A^T \Omega A = \Omega$.

Proof. We say that $A \in M(2n, \mathbb{R})$ preserves the standard symplectic form on $\mathbb{R}^{2n}$ if and only if

$$\begin{aligned} \forall x, y \in \mathbb{R}^{2n} \quad w(Ax, Ay) = w(x, y) &\iff \langle Ax, \Omega(Ay) \rangle = \langle x, \Omega y \rangle \\ &\iff \langle x, A^T \Omega Ay \rangle = \langle x, \Omega y \rangle \\ &\iff \forall y \in \mathbb{R}^{2n} A^T \Omega Ay = \Omega y \\ &\iff A^T \Omega A = \Omega. \end{aligned}$$

■

Proposition. If $A^T \Omega A = \Omega$, then A is invertible.

Proof. Suppose $A^T \Omega A = \Omega$, then we have

$$\begin{aligned}
 \det(A^T \Omega A) &= \det(\Omega) \\
 \implies \det(A^T) \det(\Omega) \det(A) &= \det(\Omega) \\
 \implies (\det(A))^2 \det(\Omega) &= \det(\Omega) \\
 \implies (\det(A))^2 &= 1 \\
 \implies \det(A) &= \pm 1 \\
 \implies A &\text{ is invertible.}
 \end{aligned}$$

■

Proposition. $A^T \Omega A = \Omega$ if and only if $-\Omega A^T \Omega = A^{-1}$ where A is invertible.

Proof. We have

$$\begin{aligned}
 -\Omega A^T \Omega = A^{-1} &\quad \Longleftrightarrow \quad -A^T \Omega = \Omega^{-1} A^{-1} \\
 &\quad \text{left multiply by } \Omega^{-1} \\
 &\quad \Longleftrightarrow \quad -A^T \Omega A = \Omega^{-1} \\
 &\quad \text{right multiply by } A \\
 &\quad \Longleftrightarrow \quad -A^T \Omega A = \Omega \\
 &\quad \Omega^{-1} = -\Omega \\
 &\quad \Longleftrightarrow \quad A^T \Omega A = \Omega.
 \end{aligned}$$

■

Example (Real Symplectic Group). $\text{Sp}(2n, \mathbb{R}) = \{A \in M(2n, \mathbb{R}) : A^T \Omega A = \Omega\}$ is a subgroup of $\text{GL}(2n, \mathbb{R})$.

Proof. We just proved that every matrix in $\text{SP}(2n, \mathbb{R})$ is invertible. So,

$$\text{SP}(2n, \mathbb{R}) \subseteq \text{GL}(2n, \mathbb{R}).$$

- We prove that $\text{SP}(2n, \mathbb{R})$ is closed under multiplication. Suppose $A, B \in \text{SP}(2n, \mathbb{R})$. We have

$$\begin{aligned}
 (AB)^T \Omega (AB) &= B^T (A^T \Omega A) B \\
 &= B^T \Omega B \\
 &= \Omega.
 \end{aligned}$$

Hence, we have $AB \in \text{SP}(2n, \mathbb{R})$.

- $\text{SP}(2n, \mathbb{R})$ contains the inverse of each of its elements. Suppose $A \in \text{SP}(2n, \mathbb{R})$. We have

$$\begin{aligned} A^T \Omega A = A &\implies (A^T)^{-1} \Omega A = (A^T)^{-1} \Omega \\ &\implies \Omega = (A^T)^{-1} \Omega A^{-1} \\ &\implies \Omega = (A^{-1})^T \Omega A^{-1} \\ &\implies A^{-1} \in \text{SP}(2n, \mathbb{R}) \end{aligned}$$

From the above, we can see that $\text{SP}(2n, \mathbb{R})$ is a subgroup of $\text{GL}(2n, \mathbb{R})$. Now, we will prove that $\text{SP}(2n, \mathbb{R})$ is a Lie subgroup of $\text{GL}(2n, \mathbb{R})$. Define the map

$$G : \text{GL}(2n, \mathbb{R}) \subseteq \mathbb{R}^{4n^2} \rightarrow M(2n, \mathbb{R}) \cong \mathbb{R}^{4n^2} \text{ by } A \mapsto A^T \Omega A - \Omega.$$

Clearly, $\text{SP}(2n, \mathbb{R}) = G^{-1}(\{0\})$. It can be shown that

$$\forall A \in \text{GL}(2n, \mathbb{R}) \quad DG(A) \text{ has rank } \frac{4n^2 - 2n}{2} = 2n^2 - n.$$

Hence, $\text{SP}(2n, \mathbb{R})$ is an embedded submanifold of $\text{GL}(2n, \mathbb{R})$ with dimension $4n^2 - (2n^2 - n) = 2n^2 + n$. Thus, $\text{SP}(2n, \mathbb{R})$ is a Lie subgroup of $\text{GL}(2n, \mathbb{R})$. ■

Definition (Homomorphisms, Isomorphisms, Automorphisms). Let G and H be two groups.

- A function $\varphi : G \rightarrow H$ is said to be a **homomorphism** if it preserves group multiplication

$$\forall g_1, g_2 \in G \quad \varphi(g_1 g_2) = \varphi(g_1) \varphi(g_2).$$

- A bijective homomorphism is called an **isomorphism**. If there is an Isomorphism between G and H , we say G and H are isomorphic, and we write $G \cong H$.
- An **endomorphism** of G is a homomorphism from G to itself.
- An **automorphism** of G is an isomorphism from G to itself.

Definition (Lie Group Homomorphism). Let G and H be two Lie groups.

- A function $f : G \rightarrow H$ is said to be a **Lie group homomorphism** if

- (i) f is a group homomorphism
- (ii) f is smooth

- A function $f : G \rightarrow H$ is said to be a **Lie group isomorphism** if

- (i) f is a group isomorphism
- (ii) f is smooth
- (iii) f^{-1} is smooth

If f is smooth and its inverse is smooth, then f is a diffeomorphism.

Example (The Determinant is a Lie Group Homomorphism). The determinant $\det : \text{GL}(n, \mathbb{R}) \rightarrow \mathbb{R}^*$ is a Lie Group Homomorphism:

- (i) $\forall A, B \in \text{GL}(n, \mathbb{R})$, we have $\det(AB) = \det(A)\det(B)$. Hence, $\det$ is a group homomorphism.
- (ii) $\det A$ is a polynomial in terms of the entries of A and so differentiable. Hence, the $\det$ is

smooth.

Similarly, the $\det GL(n, \mathbb{C}) \rightarrow \mathbb{C}^*$ is a Lie Group Homomorphism.

Example (Left Translation/right translation). Let G be a Lie Group and let $g \in G$. The **left translation** by g is the map

$$L_g : G \rightarrow G \quad h \mapsto gh.$$

The **right translation** by g is the map

$$R_g : G \rightarrow G \quad h \mapsto hg.$$

Notice that L_g is a composition of smooth maps and so it is smooth:

$$G \rightarrow G \times G \rightarrow G \quad h \mapsto (g, h) \mapsto gh.$$

Starting from left to right, we see that the first mapping is smooth because

- Fact 1: $F = (F_1, F_2)$ is smooth if and only if F_1 and F_2 are smooth.
- Fact 2: $h \mapsto g$ is a constant function implies that it is smooth.
- Fact 3: $h \mapsto h$ is the identity which is also smooth.

The second mapping $G \times G \rightarrow G$ is smooth because multiplication in G is smooth since G is a Lie group.

In fact, $L_g : G \rightarrow G$ is a diffeomorphism because the inverse L_g is $L_{g^{-1}}$:

$$\forall h \quad L_{g^{-1}} \circ L_g(h) = g^{-1}(gh) = h.$$

Unfortunately, L_g in general is NOT a group homomorphism; that is, in general

$$L_g(h_1 h_2) \neq L_g(h_1) L_g(h_2).$$

Similarly, the right translation map also satisfies the properties above but is NOT a group homomorphism.

Example (Conjugation by $g \in G$). Let G be a Lie group and let $g \in G$. The conjugation by g is the map:

$$C_g : G \rightarrow G \quad h \mapsto ghg^{-1}.$$

Note that C_g is a diffeomorphism because it is a composition of diffeomorphisms:

$$C_g = L_g \circ R_{g^{-1}}.$$

In fact, $C_g^{-1} = C_{g^{-1}}$. Also, note that

$$\forall h_1, h_2 \in G \quad C_g(h_1 h_2) = g(h_1 h_2)g^{-1} = (gh_1 g^{-1})(gh_2 g^{-1}) = C_g(h_1) C_g(h_2).$$

Hence, $C_g : G \rightarrow G$ is a Lie group isomorphism (i.e a Lie group automorphism).

3 Week 7

3.1 Kernel and Image

Definition (Kernel and Image). Let G and H be two groups. Let $f : G \rightarrow H$ be a homomorphism.

- The **kernel** of f is the set

$$\ker f = \{g \in G : f(g) = 1_H\} = f^{-1}(\{1_H\}) \subseteq G.$$

- The **image** of f is

$$\operatorname{Im} f = \{f(g) : g \in G\} = f(G) \subseteq H.$$

Both of these sets are subgroups of G and H , respectively.

Example. Consider the determinant $\det : \operatorname{GL}(n, \mathbb{R}) \rightarrow \mathbb{R}^*$ is a homomorphism. Thus,

$$\ker(\det) = \{A \in \operatorname{GL}(n, \mathbb{R}) : \det A = 1\} = \operatorname{SL}(n, \mathbb{R}).$$

3.2 A Common Theme in Mathematics

Remark (Topology). If X and Y are two topological spaces, we may consider them to be the same if there exists a function $f : X \rightarrow Y$ such that

- (1) f is bijective
- (2) f is continuous
- (3) f^{-1} is continuous

In the context of manifold theory, if M and N are two manifolds, we may consider them to be the same if there exists $f : M \rightarrow N$ such that

- (1) f is bijective
- (2) f is smooth
- (3) f^{-1} is smooth

In the context of group theory, if G and H are two groups, we may consider them to be the same if

- (1) f is bijective
- (2) f is a homomorphism

The last remark gives us an explanation of why (ii) is important.

3.3 Kernel of a Lie Group Homomorphism

Remark (Fact from Linear Algebra). Suppose A and B are two $n \times n$ matrices and A is invertible. Then

$$\operatorname{rank}(AB) = \operatorname{rank}(B) \quad \operatorname{rank}(BA) = \operatorname{rank}(B).$$

Theorem (Kernel of a Lie Group Homomorphism). Let G and H be Lie groups. Let $f : G \rightarrow H$ be a Lie group homomorphism. Then

- (i) The rank of the derivative of f is constant for all points in G ; that is, there exists $k \in \mathbb{N}$ such that for all $x \in G$, $\operatorname{rank}(Df(x)) = k$.
- (ii) $\ker(f)$ is an embedded submanifold of G of dimension $\dim(G) - k$ (and so it is a Lie subgroup of G)
- (iii) $\operatorname{Im}(f)$ can be equipped with a smooth structure that makes it a Lie subgroup of H .

Proof. Note that (ii) is a direct consequence of (i) and Fact 2 from many weeks ago. Here we will prove (i). It suffices to show that for all $x \in G$, $\operatorname{rank}(Df(x)) = \operatorname{rank}(Df(e))$ where $e = 1_G$. For all

$a, x \in G$, we have

$$\begin{aligned} f(x) &= f(aa^{-1}x) = f(a) \cdot f(a^{-1}x) \\ &= (L_{f(a)} \circ f \circ L_{a^{-1}})(x). \end{aligned}$$

By the chain rule,

$$Df(x) = DL_{f(a)}(f(a^{-1}x)) \cdot Df(a^{-1}x) \cdot DL_{a^{-1}}(x).$$

Since $L_{f(a)}$ and $L_{a^{-1}}$ are diffeomorphisms, $DL_{f(a)}$ and $DL_{a^{-1}}$ are invertible at any point. From a fact in linear algebra, for all $x, a \in G$

$$\text{rank}(Df(x)) = \text{rank}(Df(a^{-1}x)).$$

In particular, if we set $a = x$, we get

$$\text{rank}(Df(x)) = \text{rank}(Df(e))$$

which is our desired result. ■

Remark. Suppose G is a diffeomorphism. We have

$$(G^{-1} \circ G)(x) = x \implies DG^{-1}(G(x))DG(x) = I$$

Hence, $DG(x)$ is invertible.

3.4 Subgroups and Cosets

Definition (Subgroups and cosets (very important!)). Let G be a group. Given a subgroup H of G and $g \in G$, the **left coset** of H determined by g is the set

$$gH = \{gh : h \in H\}.$$

The **right coset** of H determined by g is the set

$$Hg = \{hg : h \in H\}.$$

Note that H itself is a coset for $g = 1_G$.

You need a subgroup and an element to talk about a coset.

Example ($G = (\mathbb{R}^2, +)$, $H = \text{y-axis}$ (Very important!)). Observe that $H = \{(0, y) : y \in \mathbb{R}\}$ is a subgroup of $G = (\mathbb{R}^2, +)$. Let's determine the left cosets of H . For all $(a, b) \in \mathbb{R}^2$, we have

$$\begin{aligned} (a, b) + H &= \{(a, b) + (0, y) : y \in \mathbb{R}\} = \{(a, 0 + y) : y \in \mathbb{R}\} \\ &= \{(a, y) : y \in \mathbb{R}\}. \end{aligned}$$

This represents the vertical line at $x = a$. So, in this example, G decomposes into disjoint cosets. Roughly speaking, the plane is the union of parallel lines. This organizes the mess that is $\mathbb{R}^2$ into more manageable chunks to understand. Is this by accident? NO!

Note that two distinct $g \in G$ can give us the same coset (or vertical line in our example). That is, consider the equivalent cosets

$$(1, 0) + H = (1, 1) + H = \text{vertical line at } x = 1.$$

Theorem. Let G be a group and H be a subgroup of G .

- (i) $G = \bigcup_{g \in G} gH$
- (ii) Suppose $g_1, g_2 \in G$. Then either $g_1H = g_2H$ or $g_1H \cap g_2H = \emptyset$.
- (iii) $g_1H = g_2H$ if and only if $g_1^{-1}g_2 \in H$.

Proof. (i) For all $g \in G$, clearly $gH \subseteq G$ since gH is a subgroup of G . Therefore,

$$\bigcup_{g \in G} gH \subseteq G.$$

Suppose $g' \in G$. Since $1 \in H$, we have $g'1 \in g'H$. Thus, $g' \in g'H \subseteq \bigcup_{g \in G} gH$. So, $G \subseteq \bigcup_{g \in G} gH$. Hence,

$$G = \bigcup_{g \in G} gH.$$

- (ii) It suffices to show that if $g_1H \cap g_2H \neq \emptyset$, then $g_1H = g_2H$. Suppose $g \in g_1H \cap g_2H$. Thus,

$$g = g_1h_1 = g_2h_2 \text{ for some } g_1 \text{ and } g_2 \text{ in } H.$$

We have

$$\begin{aligned} g_1 = g_2h_2h_1^{-1} &\implies g_1H = g_2h_2h_1^{-1}H \\ &\implies g_1H = g_2(h_2h_1^{-1}H) \\ &\implies g_1H = g_2H. \end{aligned}$$

- (iii) ($\Leftarrow$) Suppose $g_1^{-1}g_2 \in H$. Our goal is to show that $g_1H = g_2H$. By (ii), it suffices to show that $g_1H \cap g_2H \neq \emptyset$. We have

$$\begin{aligned} g_1^{-1}g_2 \in H &\implies g_1(g_1^{-1}g_2) \in g_1H \\ &\implies g_2 \in g_1H. \end{aligned}$$

Similarly, we have $g_2 \in g_2H$. Thus, $g_2 \in g_1H \cap g_2H$. So, we have $g_1H \cap g_2H \neq \emptyset$.

($\Rightarrow$) Assume $g_1H = g_2H$. We claim that $g_1^{-1}g_2 \in H$.

$$\begin{aligned} g_1H = g_2H &\implies g_2 \in g_1H \\ &\implies g_2 = g_1h \text{ for some } h \in H \\ &\implies g_1^{-1}g_2 = h \text{ for some } h \in H \\ &\implies g_1^{-1}g_2 \in H. \end{aligned}$$

■

Theorem (Congruence modulo H). Let G be a group and H be a subgroup. Define the relation **congruence modulo H** on G as follows:

$$g_1 \sim g_2 \iff g_1^{-1}g_2 \in H.$$

Then the above relation is an equivalence relation, and its equivalence classes are precisely the left cosets of H .

Remark. • When we look at a group G , one very natural perspective is to view it as a disjoint union of cosets of a subgroup H .

- This viewpoint organizes the elements of G and often makes it easier to extract information about the structure of the group.
- In a way, the collection of cosets is simpler and more manageable than the group itself.

Remark. • It is tempting to try to make this collection of cosets into a group, using the natural operation

$$(g_1H)(g_2H) := (g_1g_2)H.$$

However, this does not always work; in general, the operation may be ill-defined.

- The question is as follows: *Under what conditions is the above operation well-defined?*

Lemma. Let G be a group and H be a subgroup of G . The following statements are equivalent:

- (1) $\forall g \in G, \forall h \in H, ghg^{-1} \in H$
- (2) $\forall g \in G, \forall h \in H, gHg^{-1} = H$
- (3) $\forall g \in G, gH = Hg$.

3.5 Normal Subgroups

Definition (Normal Subgroups). In the case that any of the above equivalent conditions holds, we say H is a normal subgroup of G .

Notationally, normal subgroups can be represented by $H \trianglelefteq G$. If G is abelian, any subgroup of G is normal.

Theorem. Suppose G is a group and H is a subgroup of G . The operation

$$(g_1H) \cdot (g_2H) = (g_1g_2) \cdot H$$

is well-defined on the collection of left cosets of H if and only if H is a normal subgroup of G .

Proof. ($\implies$) Assume that the operation above is well-defined. We want to show that H is a normal subgroup of G . That is, it suffices to show that for all $g \in G$ and $h \in H$, we have

$$ghg^{-1} \in H.$$

That is, we want to show that (using the previous result $aH = bH \iff b^{-1}a \in H$)

$$g^{-1}H = (gh)^{-1}H.$$

That is, we want to show that

$$g^{-1}H = h^{-1}g^{-1}H \tag{*}$$

Since the coset operation is well defined, we have

$$\begin{aligned} g^{-1}H &= (1g^{-1})H = (1H)(g^{-1}H) \\ &\stackrel{1H=H=h^{-1}H}{=} (h^{-1}H)(g^{-1}H) \\ &= (h^{-1}g^{-1})H. \end{aligned}$$

($\Leftarrow$) Assume that $H \trianglelefteq G$. We want to show that the above operation is well-defined. Suppose $g_1H = g'_1H$ and $g_2H = g'_2H$. Our goal is to show that

$$(g_1g_2)H = (g'_1g'_2)H. \quad (*)$$

Since H is normal, (that is, $g'_2H = Hg'_2$ and $g'_1H = Hg'_1$)

$$\begin{aligned} g'_1g'_2H &= g'_1Hg'_2 = g_1Hg'_2 & (g'_1H &= g_1H) \\ &= g_1g'_2H & (H \trianglelefteq G \implies Hg'_2 &= g'_2H) \\ &= g_1g_2H. & (g'_2H &= g_2H) \end{aligned}$$

and we are done. ■

Theorem (Quotient Theorem for Groups). If H is a normal subgroup of G , then the operation

$$(g_1H) \cdot (g_2H) = (g_1g_2)H$$

is well-defined on cosets

The group of cosets is called the **quotient group of G by H** , and is written G/H .

Remark. The map $\pi : G \rightarrow G/H$ where $g \mapsto gH$ is a surjective homomorphism and is called the **quotient map**.

By studying quotient groups of a group which, in a sense, organizes the group's elements into simpler chunks, we can gain valuable information about the group itself. For example, we can show that if $G/Z(G)$ is cyclic, then G is abelian (where $Z(G)$ is the center of G).

The kernel of the quotient map will be a normal subgroup of G .

3.6 Making G/N a Lie Group

Remark (Observation). Suppose G is a Lie group and N is a normal algebraic subgroup of G . From this we can construct a new algebraic group G/N . In the context of Lie Theory, an important question arises:

Can we make G/N a Lie group?

It seems reasonable to look for a smooth structure on G/N that at least ensures that the quotient map $\pi : G \rightarrow G/N$ is smooth.

What is the kernel of π ? It is

$$\begin{aligned} \ker(\pi) &= \pi^{-1}\{1_GN\} = \{g \in G : \pi(g) = 1_GN\} \\ &= \{g \in G : \pi(g) = N\} \\ &= \{g \in G : gN = N\} \\ &= N. \end{aligned}$$

(very important!)

What can we conclude from this result? If we manage to find a smooth structure on G/N that ensures $\pi : G \rightarrow G/N$ is smooth, then N must be necessarily a topologically closed subset of G (recall that if π

is smooth, then it is continuous and preimages of continuous functions of closed sets are closed). If it is not closed, then there is no way to define a smooth structure on the homomorphism π .

Theorem (Quotient Theorem for Lie Groups). Let G be a Lie group, $N \trianglelefteq G$, and N is topologically closed subset of G . Then G/N can be equipped with a smooth structure that turns it into a Lie group such that the quotient map $\pi : G \rightarrow G/N$ is a surjective smooth map.

Example ($\mathbb{R}/\mathbb{Z}$). Note that $G = (\mathbb{R}, +)$ is a Lie group and $N = (\mathbb{Z}, +)$ is a subgroup of G . Since G is abelian, $N \trianglelefteq G$. Now, is N a closed subset? Yes, because the complement of $\mathbb{Z}$ (that is $\mathbb{R} \setminus \mathbb{Z}$) is open, so $\mathbb{Z}$ is a closed subset of $\mathbb{R}$. Using the Quotient theorem for Lie groups, we have that $\mathbb{R}/\mathbb{Z}$ is a Lie group.

As a matter of fact, the mapping

$$\begin{aligned} F : \mathbb{R}/\mathbb{Z} &\longrightarrow S^1 = \text{U}(1) \\ x + \mathbb{Z} &\longmapsto e^{2\pi i x} \end{aligned}$$

is a well-defined isomorphism of Lie group. And so,

$$\mathbb{R}/\mathbb{Z} \cong S^1.$$

Why is this well-defined? Suppose $x + \mathbb{Z} = y + \mathbb{Z}$. We need to show that $e^{2\pi i x} = e^{2\pi i y}$. We have

$$\begin{aligned} x + \mathbb{Z} = y + \mathbb{Z} &\implies x - y \in \mathbb{Z} \\ &\implies e^{2\pi i(x-y)} = \cos(2\pi(x-y)) + i \sin(2\pi(x-y)) \\ &\implies e^{2\pi i(x-y)} = 1 + i0 = 1 \\ &\implies e^{2\pi i x} \cdot e^{-2\pi i y} = 1 \\ &\implies e^{2\pi i x} = e^{2\pi i y}. \end{aligned}$$

Similarly, we can show that

$$\mathbb{R}^n/\mathbb{Z}^n \cong (S^1) \times \cdots \times (S^1) \quad (n\text{-dimensional torus})$$

mapped by $(x_1, \dots, x_n) + \mathbb{Z}^n \mapsto (e^{2\pi i x_1}, e^{2\pi i x_2}, \dots, e^{2\pi i x_n})$.

Example. Let $G = \text{SU}(2)$ be a Lie group, $N = \{-I, +I\}$ is a normal subgroup of $\text{SU}(2)$, then

$$\forall g \in \text{SU}(2) \quad g(\pm I)g^{-1} = \pm gg^{-1} = \pm I \in N$$

where N is closed G (N is finite subset of G). Hence, $\text{SU}(2)/\{-I, I\}$ is a Lie group. In fact,

$$\text{SU}(2)/\{-I, +I\} \cong \text{SO}(3).$$

4 Week 8

4.1 Conjugate subgroups, Conjugacy Class

Let G be a group. Recall that for each $g \in G$, the map conjugation by g , $C_g : G \rightarrow Gh \mapsto ghg^{-1}$ is a homomorphism. Since $C_g : G \rightarrow G$ is a homomorphism, for any subgroup, $H \leq G$, $C_g(H)$ will be a subgroup of G . Note that if H is a normal subgroup of G , then

$$C_g(H) = gHg^{-1} = H. \quad \forall g \in G$$

So,

$$\{C_g(H) : g \in G\} = H$$

which is just H . We can use this set as a measure of how our subgroup deviates from being normal. This motivates the following definition.

Definition (Conjugacy Class). For any subgroup $H \leq G$, we define the **conjugacy class** of H as follows:

$$\{C_g(H) : g \in G\} = \{gHg^{-1} : g \in G\}.$$

If $\tilde{H} \in \{C_g(H) : g \in G\}$, we say that subgroups H and $\tilde{H}$ are conjugate.

Example. Let $\theta \in \mathbb{R}$ be a fixed number. Consider the cyclic subgroup of $\text{SO}(3)$ generated by R_θ^z :

$$H = \langle R_\theta^z \rangle = \{R_{k\theta}^z : k \in \mathbb{Z}\}.$$

Note that for all $g \in \text{SO}(3)$, we have

$$\begin{aligned} gHg^{-1} &= \{gR_{k\theta}^zg^{-1} : k \in \mathbb{Z}\} \\ &= \text{the cyclic subgroup of } \text{SO}(3) \\ &\text{generated by rotation } \theta \text{ radians about the axis} \\ &\text{determined by the 3rd column of } g. \end{aligned}$$

Hence, we have

$$\begin{aligned} \text{Conjugacy class of } H &= \{gHg^{-1} : g \in \text{SO}(3)\} \\ &= \text{the collection of all cyclic subgroups} \\ &\text{of } \text{SO}(3) \text{ generated by rotation } \theta \text{ radians about} \\ &\text{an arbitrary fixed axis.} \end{aligned}$$

Theorem (First Isomorphism Theorem). Let $f : G \rightarrow H$ be a group homomorphism. Then $\varphi : G/\ker(f) \rightarrow \text{Im}(f)$ where $g\ker(f) \mapsto f(g)$.

Example (Observation of rotational matrices). • Rotation by an angle θ about the origin in the xy -plane can be described by the matrix

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

• Rotation by an angle θ about the z -axis in the xyz -space can be described by the matrix

$$R_\theta^z = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Example. Let ℓ be a directed line that passes through the origin in the xyz -space. In what follows, we will find a formula for rotation by an angle θ about ℓ (an arbitrary axis). Let $e_1 = (1, 0, 0)$, $e_2 = (0, 1, 0)$, $e_3 = (0, 0, 1)$ be the standard basis. Now, choose a new orthonormal basis $\{\hat{e}_1, \hat{e}_2, \hat{e}_3\}$ such that

- (1) It is right-handed ($\hat{e}_3 = \hat{e}_1 \times \hat{e}_2$)
- (2) $\hat{e}_3$ lies on ℓ (and points in the same direction as ℓ). Let $A = P_{\beta \leftarrow \hat{\beta}}$ be the change of basis matrix from $\hat{\beta}$ coordinate to β coordinates. Suppose w is a vector in $\mathbb{R}^3$. We can write

$w = w_1e_1 + w_2e_2 + w_3e_3$ and $w = \hat{w}_1\hat{e}_1 + \hat{w}_2\hat{e}_2 + \hat{w}_3\hat{e}_3$. Recall that

$$[w]_\beta = \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix}, \quad [w]_{\hat{\beta}} = \begin{pmatrix} \hat{w}_1 \\ \hat{w}_2 \\ \hat{w}_3 \end{pmatrix}.$$

4.2 Review of Linear Algebra

Suppose V and W are vector spaces with dimensions n and m , respectively. Suppose T is a linear transformation from V to W .

Remark (Huge Fact). If we choose a basis $\{b_1, \dots, b_n\}$ for V a basis $\{f_1, \dots, f_m\}$ for W , then T can be represented by an $m \times n$ matrix, denoted by $M(T)$ in the following sense: Given any vector $v = v_1b_1 + \dots + v_nb_n$ is equal to $w_1f_1 + w_2f_2 + \dots + w_mf_m$ where

$$\begin{pmatrix} w_1 \\ \vdots \\ w_m \end{pmatrix} = M(T) \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}.$$

It can be shown that

$$M(T) = (T(b_1) \quad T(b_2) \quad \dots \quad T(b_n))$$

where for each $1 \leq i \leq n$, $T(b_i)$ will be expressed in terms of the basis in $\{b_1, \dots, b_n\}$.

Now, suppose V is a vector space and $\{b_1, \dots, b_n\}$ is one basis for V and $\{\hat{b}_1, \dots, \hat{b}_n\}$ is a second basis for V . We may consider the identity map

$$I : V \rightarrow V \quad I(x) = x$$

and equip the domain with $\{\hat{b}_1, \dots, \hat{b}_n\}$ basis, and equip the codomain with $b_1, \dots, b_n$. Then the $M(I)$ is exactly the change of coordinate matrix $P_{\beta \leftarrow \hat{\beta}}$ from $\hat{\beta}$ to β . Note that $A \in \text{SO}(3)$ because the columns of A are orthonormal and A preserves orientation. Let $v = v_1e_1 + v_2e_2 + v_3e_3$ be a vector in $\mathbb{R}^3$; that is, $[v]_\beta = (v_1, v_2, v_3)$. Let w be the vector that is obtained by rotating v by θ radians about the ℓ axis. We claim that

$$[w]_\beta = AR_\theta^z A^{-1}[v]_\beta.$$

Indeed, $A^{-1}[v]_\beta = [v]_{\hat{\beta}}$. Now, in the $\hat{\beta}$ basis, the ℓ axis is the same as the z -axis. Thus, the vector w that is obtained by rotation, θ radians about ℓ will have coordinates given by

$$[w]_{\hat{\beta}} = R_\theta^z[v]_{\hat{\beta}} = R_\theta^z A^{-1}[v]_\beta.$$

Therefore,

$$[w]_\beta = A[w]_{\hat{\beta}} = AR_\theta^z A^{-1}[v]_\beta.$$

That is,

$$[w]_\beta = (AR_\theta^z A^{-1})[v]_\beta.$$

4.3 Remark about First Isomorphism Theorem

Suppose G is a group and K is a normal subgroup of G . Then, as we know, the quotient map $\pi : G \rightarrow G/K$ where $g \mapsto gK$ is a surjective homomorphism. The above theorem for groups tells us that any surjective homomorphism with domain G is of the above form. That is, if $f : G \rightarrow H$ is a surjective homomorphism, we have

$$H \cong G/\text{Ker}(f).$$

Hence, your surjective homomorphism is, indeed, $G \rightarrow G/\text{Ker}(f)$. This theorem is giving us a way to construct all homomorphisms with domain G . In fact, we can make a connection to the following remarkable result from Category Theory.

4.4 Consequence of Yoneda Lemma

If you want to understand a mathematical object X , it suffices to understand all the structure preserving maps with domain X .

4.5 Second Isomorphism Theorem

Let G be a group. Let $K \trianglelefteq G$. As we know, the quotient map $\pi : G \rightarrow G/K$ is a surjective homomorphism. Suppose H is a subgroup of G . What is exactly $\pi(H)$ (we know that it must be a subgroup of G/K)? By the correspondence theorem, we know that

$$\pi(H) = (\text{a subgroup of } G \text{ that contains } K)/K.$$

The answer is:

$$\pi(H) = HK/K.$$

Theorem (Second Isomorphism Theorem). Let G be a group, H is a subgroup of G , K is a normal subgroup of G , then

$$\varphi : HK/K \rightarrow H/(H \cap K), \quad hk \mapsto h(H \cap K)$$

is an isomorphism

4.6 Third Isomorphism Theorem

Let G be a group, according to the first isomorphism theorem, in order to study homomorphisms with domain G , it suffices to study quotients G/N where N is a normal subgroup of G . Will it be helpful now to consider the quotients of G/N ? No! There will be no new information if you construct the quotients of G/N .

Theorem (Third Isomorphism Theorem). Let G be a group, K be a normal subgroup of G , N is a subgroup of K , and N is a normal subgroup of G . Then

$$\varphi : (G/N)/(K/N) \rightarrow G/K, \quad gN(K/N) \mapsto gK$$

is an isomorphism.

4.7 The Correspondence Theorem

Let G be a group and K be a normal subgroup of G . There are two questions we need to answer.

- (1) What are the subgroups of G/K ?
- (2) What are the normal subgroups of G/K ?

The answer to these questions are the following:

- (1) There is a one-to-one correspondence between the set of all subgroups of G that contain K and the set of all subgroups of G/K :
 - (i) If $H \leq G$ and $R \leq G/K$, then $H/K \leq G/K$.
 - (ii) If $R \leq G/K$, then $R = H/K$ for some $H \leq G$ and $K \subseteq H$.
 - (ii) The above correspondence preserves normality:

$$H/K \trianglelefteq G/K \iff H \trianglelefteq G, K \subseteq H.$$

4.8 Simple Groups

Definition (Simple Groups). Let G be an algebraic group. We say that G is **simple** if

- (1) $G \neq \{1_G\}$
- (2) The only normal subgroups of G are G itself and $\{1_G\}$.

Cannot make the group much simpler; that is, we cannot quotient the group out by a normal subgroup even further.

Remark (Observation). Many-to-one functions help us simplify/organize the domain of the function. An example is the quotient map $\pi : G \rightarrow G/N$ defined by $g \mapsto gN$. This organizes the domain of π into cosets.

Remark. The following two statements are equivalent:

- (1) G is not simple
- (2) There exists a many-to-one homomorphism of G onto a group G' other than the homomorphism sending all elements to 1.

That is, G' can be viewed as a simpler version of G (or, at any rate, not more complicated than G).

Remark. • A **simple Lie group** is not usually defined as a Lie group that is simple in the sense of abstract group theory.

- The most common definition is that a Lie group G is called **simple** if its corresponding Lie algebra is a simple Lie algebra.
- As we shall see, with this definition, a simple Lie group may not be simple as an abstract group.

4.9 Direct Product of Groups

Definition (Direct Product of Groups). If G_1 and G_2 are two groups, the **direct product** of G_1 and G_2 is the set $G_1 \times G_2$ with the group structure defined by the following multiplication rule:

$$(g_1, g_2) \cdot (g'_1, g'_2) = (g_1 g'_1, g_2 g'_2)$$

and the identity is

$$1_{G_1 \times G_2} = (1_{G_1}, 1_{G_2}).$$

If G_1 and G_2 are Lie groups, then $G_1 \times G_2$ is a Lie group.

- (1) $G_1 \times G_2$ is an algebraic group (previous slide)
- (2) G_1 is a smooth manifold and G_2 is a smooth manifold, from a month ago, we have $G_1 \times G_2$ is a smooth manifold.
- (3) multiplication map: $(g_1, g_2) \cdot (g'_1, g'_2) = (g_1 g'_1, g_2 g'_2)$. The multiplication in the first and second components are smooth operations. Hence, the multiplication in the direct product is also smooth.
- (4) Inversion: $(g_1, g_2) \mapsto (g_1^{-1}, g_2^{-1})$.

Example. We know that $S^1 = U(1) = \text{SO}(2)$ is a Lie group. It is, indeed, a compact and connected.

Thus, $\underbrace{S^1 \times \cdots \times S^1}_{k \text{ times}}$ is also a Lie group. Note that this set is

- (1) connected
- (2) compact
- (3) abelian

Why is S^1 closed? We know that $S^1 = f^{-1}(1)$ where $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x, y) = x^2 + y^2$ and that because f is continuous, we know that the preimage of the closed set $\{1\}$ is closed. Clearly, we also see that S^1 is bounded. Since each set is compact, and there are k number of them, we see that because there are a finite product of compact sets, we know that the finite product must also be compact. Note that it is also true that a finite product of connected topological spaces is connected.

In fact, the only k -dimensional, where $k \geq 1$, Lie group that is compact, connected, and abelian is T^k . That is, if G is a k -dimensional compact connected abelian Lie group, then $G \cong T^k$ (where $\cong$ is a Lie group homomorphism). Here are some core ideas of the proof:

- (1) Use the fact G is abelian to show that the Lie Algebra of G is isomorphic to $\mathbb{R}^k$ with $[x, y] = 0$.
- (2) Use the facts that G is compact and connected to show that

$\exp : g \rightarrow G$ is surjective.

- (3) Use the fact that G is abelian to show that $\exp : g \rightarrow G$ is indeed a group homomorphism; that is, $g/\ker(\exp) \cong G$
- (4) Use the fact that G is compact and that $g/\ker(\exp) \cong \mathbb{Z}^k$.
- (5) Conclude that

$$G \cong g/\ker(\exp) \cong \mathbb{R}^k/\mathbb{Z}^k \cong (S^1)^k = T^k.$$

End of Week 8

5 Week 9

In mathematics it is often helpful, when studying an object X (such as an algebraic structure, a metric space, or a normed space) to identify certain subsets/subobjects/substructures $E \subseteq X$ that are easier to work with or possess nice properties.

Example (Analysis). $X = C([0, 1], \mathbb{R})$ and $E = \mathbb{P}$. The following fact is true:

$\forall f \in X$ there exists $(P_n) \subseteq E$ such that $p_n \rightarrow f$ uniformly.

Example (Linear Algebra). Let $X = M(n, \mathbb{R})$ and $E =$ Diagonal $n \times n$ real matrices. The connection here is that if $A \in X$ has n linearly independent eigenvectors $A = PDP^{-1}$ where P is invertible and $D \in E$. Another connection is that if $A \in X$ is symmetric, then $A = PDP^T$ where P is orthogonal and $D \in E$. (We say that every symmetric matrix is orthogonally diagonalizable)

Example (Complex Matrices). Let $X = M(n, \mathbb{C})$ and $E =$ diagonal $n \times n$ matrices in X . If A is unitary (Hermitian) in X , then $PDP^* = A$ for some unitary matrix P and $D \in E$.

Remark (Linear Algebra). If $A \in M(n, \mathbb{C})$, then A is unitarily diagonalizable if and only if $A^*A = 0$.

AA^* where such matrices are called normal matrices. This tells us that hermitian and unitary matrices are normal in this case.

Definition (Torus Subgroup). Let G be a Lie group. A Lie subgroup H of G is said to be a **Torus subgroup** of dimension $k \geq 1$ if

$$H \cong T^k$$

where $\cong$ is a Lie group isomorphism. That is, H is a k -dimensional Torus subgroup if H has dimension k and it is compact, connected, and abelian.

Definition (Maximal Torus). Let G be a Lie group and H be a Torus subgroup of G . We say that H is a **maximal torus** if H is not contained in any larger Torus subgroup of G .

Example $(\mathrm{SO}(2) = \mathrm{U}(1) = S^1)$. Clearly, $\mathrm{SO}(2)$ is its own maximal Torus.

Example $(\mathrm{SO}(3))$. Note that

$$H = \left\{ R_\theta^z = \begin{pmatrix} \cos \theta & \sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} : \theta \in \mathbb{R} \right\}$$

is a copy of $\mathrm{SO}(2) = S^1 = T^1$ inside $\mathrm{SO}(3)$, so it is a Torus subgroup. We claim that H is maximal torus in $\mathrm{SO}(3)$. The reason is as follows: Let T be a Torus subgroup of $\mathrm{SO}(3)$ that contains H . It suffices to show that $T = H$ (and also H is maximal). Since any Torus subgroup is abelian, any $A \in T$ commutes with all $R_\theta^z \in H$:

$$\forall A \in T \text{ and } \forall \theta \in \mathbb{R} \quad AR_\theta^z = R_\theta^z A.$$

In what follows, we will show that if $A \in \mathrm{SO}(3)$ satisfies (*), then $A \in H$. Let e_1, e_2, e_3 be our standard basis vectors. In order to show that $A \in \mathrm{SO}(3)$ that satisfies (*) is indeed H , it suffices to show that:

- (1) $A|_{(e_1, e_2)\text{-plane}}$ is a Linear isometry of the (e_1, e_2) plane.
- (2) $\det A > 0$.

Proof. (1) It is enough to show that $Ae_1 \in (e_1, e_2)$ -plane and $Ae_2 \in (e_1, e_2)$ -plane. Let $Ae_1 = \alpha_1 e_1 + \alpha_2 e_2 + \alpha_3 e_3$. Our goal is to show that $\alpha_3 = 0$. Note that, by assumption, A commutes with all R_θ^z , in particular, it commutes with

$$R_\pi^z = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Therefore,

$$AR_\pi^z(e_1) = R_\pi^z A(e_1) \iff AR_\pi^z(e_1) = A$$

■

Main point of this lecture is that not all Lie groups have a Torus subgroup.

End of Tuesday Lecture

Remark. We can travel from the Lie Algebra to the Lie group using the $\exp : \text{Lie}(G) \rightarrow G$ (where $\text{Lie}(G)$ is a Lie Algebra of the Lie group). This tells us that every compact Lie group has a Torus subgroup. Indeed, if x is any nonzero vector in $\text{Lie}(G)$, then the closure of $\{\exp(tx) : t \in \mathbb{R}\}$ is a compact connected abelian Lie subgroup of G . Note that H is a subgroup of $\text{Lie}(G)$ which is based on the two observations:

- (1) $e^{tx}e^{sx} = e^{(t+s)x} = e^{sx} \cdot e^{tx}$
- (2) $e^{tx}e^{-tx} = e^{0x} = 1_G$.

Note that

- (1) If H is a subgroup of Lie group G , then $\overline{H}$ is also a subgroup of G . Furthermore, if this is also abelian (which it is), then the closure is also abelian.
- (2) Closure of a connected set is connected (in any topological space).
- (3) A closed subset of a compact is compact.

We will prove (1) in this case.

Let G be a Lie group. Let H be an algebraic subgroup of G . Then the closure of H , denoted by $\overline{H}$, is a closed subgroup of G (and so it is a Lie subgroup of G). Moreover, if H is abelian, $\overline{H}$ is abelian also.

Proof. Note that clearly, $\overline{H}$ is closed. It suffices to show that $\overline{H}$ is closed under multiplication. Suppose a and b are in $\overline{H}$. Our goal is to show that $ab \in \overline{H}$. Note that $a \in \overline{H}$, there exists a sequence (a_n) in H such that $a_n \rightarrow a$ in G . Since $b \in \overline{H}$, there exists a sequence (b_n) in H such that $b_n \rightarrow b$ in G . Therefore,

$$(a_n, b_n) \rightarrow (a, b) \text{ in } G \times G.$$

Let's denote multiplication in G by $m : G \times G \rightarrow G$. We know $m : G \times G \rightarrow G$ is continuous. Then

$$m(a_n, b_n) = m(a, b) \implies a_n b_n \rightarrow ab \in G.$$

So, ab is the limit of a sequence of H . Thus, $ab \in \overline{H}$.

Also, $\overline{H}$ contains the inverse of each of its elements. Suppose $a \in \overline{H}$. Our goal is to show that $a^{-1} \in \overline{H}$. We have

$$a \in \overline{H} \implies \exists (a_n) \subseteq H \text{ such that } a_n \rightarrow a \in G.$$

Since inversion is a continuous map, we have

$$a_n^{-1} \rightarrow a^{-1} \in G.$$

Thus, a^{-1} is the limit of a sequence in H . Thus, $a^{-1} \in \overline{H}$.

Now, we want to prove that $\overline{H}$ is abelian. Suppose $a, b \in \overline{H}$. Our goal is to show that $ab = ba$. Similarly to what we saw above, there exists $(a_n) \subseteq H$ such that $a_n \rightarrow a$ in G and there exists a sequence $(b_n) \subseteq H$ such that $b_n \rightarrow b$ in G . So, we have $a_n b_n \rightarrow ab$ and $b_n a_n \rightarrow ba$. Then $ab = ba$. ■

Remark (Fact Used in the Above Proof From Analysis). If $z_n \rightarrow z$ and f is continuous, then $f(z_n) \rightarrow f(z)$.

6 Week 10

6.1 Tangent Spaces

Let M be a smooth manifold. What do we mean by a **smooth path** in M ?

Definition (Smooth Path). A **smooth path** in M is a smooth function $f : I \rightarrow M$ with $t \mapsto f(t)$ where $I \subseteq \mathbb{R}$ is a nonempty open interval. (We refer to this path as the path f or the path $f(t)$).

- Each component of the function is a smooth path in $\mathbb{R}$.
- Path connected implies continuous paths between two points a and b .
- We can also extend this notion to smooth paths between a and b .

Example (Examples of Smooth Paths). • $r : (-1, 1) \rightarrow \mathbb{R}^2$ defined by $r(t) = (t, t^2)$ is a smooth path in $M = \mathbb{R}^2$.

- $A : (0, \pi) \rightarrow \text{SO}(2)$, where

$$A(t) = \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}$$

is a **smooth path** in $M = \text{SO}(2)$. Consider the following two situations: Let $f : (0, \pi) \rightarrow \mathbb{R}^4$ where $t \mapsto (\cos t, -\sin t, \sin t, \cos t)$ and $A : (0, \pi) \rightarrow \text{SO}(2)$ where

$$t \mapsto \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}.$$

Note from these two examples, the implicit assumption is $\text{SO}(2) \subseteq \text{GL}(2, \mathbb{R}) \subseteq \mathbb{R}^4$.

- (An example of a non-smooth map) Let $A : (-\pi/2, \pi/2) \rightarrow \text{SO}(2)$, where

$$A(t) = \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}$$

is not a smooth path in $\text{SO}(2)$. (However, it is a continuous path in $\text{SO}(2)$).

Theorem (Manifold Theory (Relating to the 2nd Example Above)). Suppose N, M, L are three smooth manifolds. Suppose M is an embedded sub-manifold of L . Suppose $F : N \rightarrow M$. Then $F : N \rightarrow M$ is smooth if and only if $i \circ F : N \rightarrow L$ where i is the inclusion map ($i : M \rightarrow L$)

In our example (2), $N = (0, 1)$ and $M = \text{SO}(2)$ and $L = \mathbb{R}^4$. It is clear that $A : N \rightarrow L$ is smooth (because each smooth was smooth). So, by the above claim $A : N \rightarrow M$ is also smooth.

Note that derivative of a path can be thought of in the following way:

- If $r(t) = (r_1(t), r_2(t), \dots, r_n(t))$ is a smooth path in $\mathbb{R}^n$, then

$$r'(t) = \lim_{\delta t \rightarrow 0} \frac{r(t + \delta t) - r(t)}{\delta t} = (r'_1(t), r'_2(t), \dots, r'_n(t)).$$

Remark. Let M be an embedded submanifold of $\mathbb{R}^k$. Let $p \in M$. As we know the tangent space to M at p , $T_p M$, can be viewed as a vector space associated with $p \in M$. What is the precise definition of $T_p M$?

The answer is: There are multiple ways to define $T_p M$. Here is one of them

- $v \in T_p M$ if and only if there exists a path $A(t)$ in M such that $A(0) = p$ and $A'(0) = v$.

- More precisely, $v \in T_p M$ if and only if there exists $\varepsilon > 0$ and a smooth path $A : (-\varepsilon, \varepsilon) \rightarrow M$ such that $A(0) = p$ and $A'(0) = v$.
- Note that $A'(0)$ is sometimes referred to as the velocity vector of the path $A(t)$ at the point p .

Note that $T_p M$ consists of the velocity vectors of all smooth paths in M through $p \in M$.

Example. Prove that

$$(1) \quad T_I(\mathrm{O}(2)) = \{A \in M(2, \mathbb{R}) : A^T = -A\} = \left\{ \begin{pmatrix} 0 & -\theta \\ \theta & 0 \end{pmatrix} : \theta \in \mathbb{R} \right\}.$$

$$(2) \quad T_I(\mathrm{SO}(2)) = \{A \in M(2, \mathbb{R}) : A^T = -A\}.$$

Proof. Here we will prove (2). The proof of (1) is the same.

- (I) We claim that $T_I(\mathrm{SO}(2)) \subseteq \mathrm{skew}(2, \mathbb{R})$. Let $X \in T_I(\mathrm{SO}(2))$. This tells us that there exists a smooth path $A : (-\varepsilon, \varepsilon) \rightarrow \mathrm{SO}(2)$ such that $A(0) = I$ and $A'(0) = X$. Note that for all $t \in (-\varepsilon, \varepsilon)$ $A(t) \in \mathrm{SO}(2)$. Then for all $t \in (-\varepsilon, \varepsilon)$ $A(t)^T A(t) = I$. Hence, we have

$$\frac{d}{dt}[A(t)^T A(t)] = 0 \implies A'(t)^T A(t) + A(t)^T A'(t) = 0.$$

In particular, if $t = 0$, we get

$$\begin{aligned} A'(0)^T A(0) + A(0)^T A'(0) &= 0 \\ \implies X^T I + I^T X &= 0 \\ \implies X^T + X &= 0 \\ \implies X^T &= -X. \end{aligned}$$

Hence, $X \in \mathrm{Skew}(2, \mathbb{R})$.

- (II) Now, we show that $\mathrm{Skew}(2, \mathbb{R}) \subseteq T_I \mathrm{SO}(2)$. Let $\begin{pmatrix} 0 & -\theta \\ \theta & 0 \end{pmatrix}$ be an arbitrary elements of $\mathrm{Skew}(2, \mathbb{R})$. We need to show that this matrix is in $T_I \mathrm{SO}(2)$. That is, we need to show that there exists a smooth path $A(t)$ in $\mathrm{SO}(2)$ such that $A(0) = I$ and $A'(0) = \begin{pmatrix} 0 & -\theta \\ \theta & 0 \end{pmatrix}$. Consider the path $A : (-1, 1) \rightarrow \mathrm{SO}(2)$ given by

$$A(t) = \begin{pmatrix} \cos(\theta t) & -\sin(\theta t) \\ \sin(\theta t) & \cos(\theta t) \end{pmatrix}.$$

Notice that, clearly, $A(0) = I$. Also,

$$A'(t) = \begin{pmatrix} -\theta \sin(\theta t) & -\theta \cos(\theta t) \\ \theta \cos(\theta t) & -\theta \sin(\theta t) \end{pmatrix}$$

and so we have

$$A'(0) = \begin{pmatrix} 0 & -\theta \\ \theta & 0 \end{pmatrix}.$$

■

Remark. (i) As we saw in part 1 of the proof above, one can often find a condition that tangent vectors at the identity of a classical matrix Lie group G must satisfy by differentiating the defining equation of G .

- (ii) Once this condition is identified, the next question is whether every vector satisfying it actually arises as a tangent vector at the identity. In other words, we need to check whether there exists

smooth paths in G whose tangent at the identifies matches the condition (this was part 2 of our solution).

- (iii) It is at this stage, when constructing smooth paths in G , that a particularly useful tool, **the matrix exponential**, comes to our aid.

Definition (Matrix Exponential). Let $A \in M(n, \mathbb{C})$. We define the exponential of A , denoted by $\exp(A)$ or e^A , as follows:

$$\exp(A) = \sum_{m=0}^{\infty} \frac{A^m}{m!} = I + A + \frac{A^2}{2!} + \cdots$$

Example. • Let $D = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$. Compute e^D . Recall that if D is any diagonal matrix, then for every natural number m , D^m is just raising the diagonal entries by m . Then

$$\begin{aligned} e^D &= \sum_{m=0}^{\infty} \frac{D^m}{m!} = \sum_{m=0}^{\infty} \frac{1}{m!} \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}^m \\ &= \sum_{m=0}^{\infty} \frac{1}{m!} \begin{pmatrix} 2^m & 0 \\ 0 & 3^m \end{pmatrix} \\ &= \end{aligned}$$

For the next step, let us be careful when distributing the series in. Note that

$$\sum_{m=0}^{\infty} \frac{1}{m!} \begin{pmatrix} 2^m & 0 \\ 0 & 3^m \end{pmatrix} = \lim_{N \rightarrow \infty} \sum_{m=0}^N \frac{1}{m!}$$

7 Week 11

7.1 Opening Remarks with Exponential of a Matrix

Remark. For which matrices is it easy to compute the exponential?

- (1) Diagonal matrices
- (2) Nilpotent matrices (finite sum); that is, matrices where there exists a $k \in \mathbb{N}$ such that $A^k = O_{n \times n}$.

- An example,

$$A = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$$

is nilpotent.

- More generally, nilpotent 3×3 matrices have the following form

$$\begin{pmatrix} a & a & a \\ b & b & b \\ (a+b) & -(a+b) & -(a+b) \end{pmatrix}$$

is nilpotent.

- (3) Diagonalizable matrices

Proposition (Powers and Exponential of Diagonalizable Matrices). Suppose that $A \in M(n, \mathbb{C})$ is **Diagonalizable**, that is, suppose there exists an invertible matrix P and a diagonal matrix D such that

$$A = PDP^{-1}.$$

Then

- (i) For all $m \in \mathbb{N}$ $A^m = PD^m P^{-1}$.
- (ii) $\exp(A) = P(\exp(D))P^{-1}$

Proof. (i) For $m = 1$, $A^1 = PD^1 P^{-1}$. For $m = 2$,

$$A^2 = AA = PDP^{-1}(PDP^{-1}) = PDDP^{-1} = PD^2 P^{-1}.$$

We can do a similar argument by using induction.

(ii) By definition, we have

$$\begin{aligned} e^A &= \sum_{m=0}^{\infty} \frac{A^m}{m!} = \sum_{m=0}^{\infty} \frac{(PDP^{-1})^m}{m!} \\ &= \sum_{m=0}^{\infty} \frac{PD^m P^{-1}}{m!} \\ &= P \left(\sum_{m=0}^{\infty} \frac{D^m}{m!} \right) P^{-1} \\ &= Pe^D P^{-1}. \end{aligned}$$

■

Remark (Review of Linear Algebra). If the matrix $A \in M(n, \mathbb{C})$ has n linearly independent eigenvectors, then A is diagonalizable. Indeed,

$$A = PDP^{-1}$$

where

- (*) Columns of P are linearly independent eigenvectors of A
- (*) Diagonal entries of D are the corresponding eigenvalues.

7.2 Review of Eigenvectors

Example. If $A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$, the eigenvalues are 4 and 2 which correspond to the (linearly independent) vectors $(1, 1)$ and $(-1, 1)$. Since the number of linearly independent vectors is 2. Then

$$A = PDP^{-1} \text{ where } P = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \text{ and } D = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}$$

where

$$A \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 4 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

and

$$A \begin{pmatrix} -1 \\ 1 \end{pmatrix} = 2 \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

Also, it is true that

$$A = PDP^{-1}$$

where

$$P = \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}, D = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}.$$

Example. Let

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ 3 & 3 & 4 \end{pmatrix}$$

where the eigenvalues and the corresponding eigenvectors are 1 and 7 with $(-1, 1, 0)$ and $(-1, 0, 1)$ (corresponding to 1) and 7 corresponding to $(1, 2, 3)$. Then since there are 3 linearly independent eigenvectors, we have

$$P = \begin{pmatrix} -1 & -1 & 1 \\ 1 & 0 & 2 \\ 0 & 1 & 3 \end{pmatrix}$$

and

$$D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 7 \end{pmatrix}.$$

In summary, in order to compute e^X where X is diagonalizable requires the following steps:

- (1) Write $X = PDP^{-1}$
- (2) Then $e^X = Pe^DP^{-1}$.

7.3 Computing Exponential of Matrices

Example (Skew Symmetric Matrices and its Exponential). Let $\theta \in \mathbb{R}$ be a fixed number. Let $X = \begin{pmatrix} 0 & -\theta \\ \theta & 0 \end{pmatrix}$. Prove that

$$e^X = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

Proof. The corresponding eigenvalues and eigenvectors of this matrix are $-i\theta$ corresponding to $(1, i)$ and $i\theta$ corresponding to $(i, 1)$. Check this in private! So, $X = PDP^{-1}$ where

$$P = \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}, D = \begin{pmatrix} -i\theta & 0 \\ 0 & i\theta \end{pmatrix}.$$

Note that

$$P^{-1} = \frac{1}{1-i^2} \begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{-i}{2} \\ \frac{-i}{2} & \frac{1}{2} \end{pmatrix}.$$

Therefore,

$$e^X = Pe^DP^{-1} = \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} \begin{pmatrix} e^{-i\theta} & 0 \\ 0 & e^{i\theta} \end{pmatrix} \begin{pmatrix} \frac{1}{2} & \frac{-i}{2} \\ \frac{-i}{2} & \frac{1}{2} \end{pmatrix}.$$

Then we have

$$\begin{aligned} e^X &= \begin{pmatrix} e^{-i\theta} & ie^{i\theta} \\ ie^{-i\theta} & e^{i\theta} \end{pmatrix} \begin{pmatrix} \frac{1}{2} & \frac{-i}{2} \\ \frac{-i}{2} & \frac{1}{2} \end{pmatrix} \\ &= \begin{pmatrix} \frac{e^{-i\theta} + e^{i\theta}}{2} & \frac{-ie^{-i\theta} + ie^{i\theta}}{2} \\ \frac{ie^{-i\theta} - ie^{i\theta}}{2} & \frac{e^{-i\theta} + e^{i\theta}}{2} \end{pmatrix} \\ &= \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \end{aligned}$$

where

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2} \text{ and } \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}.$$

Therefore, we have if we have $X = \begin{pmatrix} 0 & -\theta \\ \theta & 0 \end{pmatrix}$, then

$$e^{tx} = \begin{pmatrix} \cos(\theta t) & -\sin(\theta t) \\ \sin(\theta t) & \cos(\theta t) \end{pmatrix} \quad \forall t \in \mathbb{R}.$$

■

Remark. In general, every matrix $X \in M(n, \mathbb{C})$ can be written (uniquely) as

$$X = S + N$$

where S is diagonalizable, N is nilpotent, and $SN = NS$. Then

$$e^X = e^{S+N} = e^S e^N$$

where e^S can be easily computed by diagonalization and e^N can be computed by converting the infinite sum to a finite sum.

Remark. Let G be a Lie group and $e \in G$, be its identity element. Without loss of generality, we may assume that G is an embedded submanifold of $\mathbb{R}^k$ for some $k \in \mathbb{N}$ (so, our construction of $T_e G$ is valid). A question we can ask is "what sort of algebraic operations can we define on $T_e G$?"

7.4 Operations of Tangent Space at the identity of a Lie Group

Let $X, Y \in T_e G$. Then there exists a smooth path $A(t)$ in G such that $A(0) = e$ and $A'(0) = X$. Likewise, there exists another path $B(t)$ in G such that $B(0) = e$ and $B'(0) = Y$. The first immediate operation we can do with these two paths is to multiply them together which in of itself is a smooth path because the operation is smooth. That is,

$$C(t) = A(t)B(t)$$

is another smooth path in G with $C(0) = e$ and so we have $C'(0) \in T_e G$. In particular, if G is a matrix Lie group, then

$$C'(t) = A'(t)B(t) + A(t)B'(t) \implies C'(0) = A'(0)B(0) + A(0)B'(0) = XI + IY = X + Y.$$

If X and Y are in $T_l G$, then $X + Y \in T_l G$. Hence, we now know that $T_e G$ is closed under addition.

Now, we will show that this space is closed under scalar multiplication. For any $\alpha \in \mathbb{R}$, $C(t) = A(\alpha t)$ is another smooth path in G with $C(0) = e$. Then $C'(0) \in T_e G$. In particular, if G is a matrix Lie group, then

$$\begin{aligned} C'(t) &= \alpha A'(\alpha t) \\ &= \alpha A'(0) \\ &= \alpha X. \end{aligned}$$

So, we proved that if $X \in T_e G$ and $\alpha \in \mathbb{R}$, then $\alpha X \in T_e G$. Hence, this tells us that $T_e G$ is a linear subspace of $\mathbb{R}^k$.

Note that $C(t) = [A(t)]^{-1}$ is another smooth path in G with $C(0) = e$. That is, $C'(0) \in T_e G$. In particular, if G is matrix Lie group. Then

$$\begin{aligned} C'(t) &= -A^{-1}(t)A'(t)A^{-1}(t) \\ \implies C'(0) &= -A^{-1}(0)A'(0)A^{-1}(0) \\ &= -IXI \\ &= -X. \end{aligned}$$

That is, if $X \in T_I G$, then $-X \in T_I G$ (a special case of the last proof).

Next, if we are given $E \in G$ where $C(t) = EB(t)$ is a smooth path in G but $c(0) \neq e$. However, we may consider $C(t) = EB(t)E^{-1}$. Then $C(0) = e$. Indeed,

$$C(0) = EB(0)E^{-1} = EeE^{-1} = EE^{-1} = e.$$

In particular, if G is a matrix Lie group, then

$$\begin{aligned} C'(t) &= EB'(t)E^{-1} \\ \implies C'(0) &= EYE^{-1}. \end{aligned}$$

That is, if $Y \in T_I G$, then for every matrix $E \in G$, $EYE^{-1} \in T_I G$.

For all s , $C_s(t) = A(s)B(t)A(s)^{-1}$ is a smooth path in G and for all s , $C_s(0) = e$. Then for all s , $C'_s(0) \in T_e G$. So, if we let $v(s) = C'_s(0)$, then $v(s)$ is a path inside $T_e G$. Thus, in particular, $v'(0) \in T_e G$. That is,

$$\left. \frac{d}{ds} \right|_{s=0} \left. \frac{\partial}{\partial t} \right|_{t=0} A(s)B(t)A^{-1}(s) \in T_e G.$$

Note that please notice that if G is abelian, then the above expression is 0. Now, suppose G is a matrix Lie group. Let's compute the above vector in $T_e G$. We have

$$\begin{aligned} \left. \frac{d}{ds} \right|_{s=0} \left. \frac{\partial}{\partial t} \right|_{t=0} A(s)B(t)A^{-1}(s) &= \left. \frac{d}{ds} \right|_{s=0} A(s)B'(0)A^{-1}(s) \\ &= A'(0)B'(0)A^{-1}(0) + A(0)B'(0)(-A(0)A'(0)A^{-1}(0)) \\ &= A'(0)B'(0)A^{-1}(0) - A(0)B'(0)(A(0)A'(0)A^{-1}(0)) \\ &= XYI - IY(IXI) \\ &= XY - YX. \end{aligned}$$

This remarkable equation that we have obtained at the end is called the **Lie Bracket**.

Remark (Observation). Suppose G is an abstract group. Note that G is abelian if and only if for all $x, y \in G$ $xy = yx$. This is equivalent to for all $x, y \in G$, $(xy)(yx)^{-1} = e$ which is further equivalent to for all $x, y \in G$, $xy(x^{-1}y^{-1}) = e$.

Similarly, for all s , $C_s(t) = A(s)B(t)A^{-1}(s)B^{-1}(t)$ is a smooth path in G and for all s , $C_s(0) = e$. Then we have for all s , $C'_s \in T_e G$. Therefore, if we define $v(s) = C'_s(0)$, then $v(s)$ is a path inside the vector space $T_e G$. Thus, $v'(0) \in T_e G$. That is,

$$\left. \frac{d}{ds} \right|_{s=0} \left. \frac{d}{dt} \right|_{t=0} A(s)B(t)A^{-1}(s)B^{-1}(t) \in T_e G$$

Note that if G is abelian, the above expression is zero. In particular, if G is a matrix Lie group, we have

$$\begin{aligned}
 \left. \frac{d}{ds} \right|_{s=0} \left. \frac{\partial}{\partial t} \right|_{t=0} [A(s)B(t)A^{-1}(s)B^{-1}(t)] &= \left. \frac{d}{ds} \right|_{s=0} [A(s)B'(t)A^{-1}(s)B^{-1}(t) + A(s)B(t)A^{-1}(s)(-B^{-1}(t)B'(t)B^{-1}(t))] \Big|_{t=0} \\
 &= \left. \frac{d}{ds} \right|_{s=0} [A(s)B'(0)A^{-1}(s)B^{-1}(0) \\
 &\quad - A(s)B(0)A^{-1}(s)B^{-1}(0)B'(0)B^{-1}(0)] \\
 &= \left. \frac{d}{ds} \right|_{s=0} [A(s)YA^{-1}(s) - A(s)A^{-1}(s)Y] \\
 &= \left. \frac{d}{ds} \right|_{s=0} [A(s)YA^{-1}(s) - Y] \\
 &= \left. \frac{d}{ds} \right|_{s=0} [A(s)YA^{-1}(s)] \\
 &= XY - YX
 \end{aligned}$$

(See previous proof)

Remark. Note that from the previous two cases produce the zero map on $T_e G$ if G is abelian:

$$\forall x, y \in T_e G \quad (x, y) \mapsto 0.$$

We define the **Lie Bracket** on $T_e G$ by for all $X, Y \in T_e G$

$$[X, Y] = \left. \frac{d}{ds} \right|_{s=0} \left. \frac{\partial}{\partial t} \right|_{t=0} A(s)B(t)A^{-1}(s)B^{-1}(t).$$

When we equip $T_e G$ with this Lie bracket is called the Lie algebra of g and is denoted by $\text{Lie}(G)$.

7.5 Lie Algebras

Definition (Lie Algebra). A real (or complex) Lie algebra is a real (or complex) vector space L equipped with a function $[\cdot, \cdot] : L \times L \rightarrow L$.

- (i) $[\cdot, \cdot]$ is bilinear
- (ii) $[\cdot, \cdot]$ is skew-symmetric
- (iii) $[\cdot, \cdot]$ satisfies the Jacobi identity; that is, for all $x, y, z \in L$, we have

$$[x, [y, z]] + [z, [x, y]] + [y, [z, x]] = 0.$$

One way to create a Lie Algebra is to start with a Lie group and look at the tangent space at the identity of the Lie group and define a Lie Bracket on it.

7.6 Main Properties Of The Exponential

Proposition (Properties). Let X and Y be in $M(n, \mathbb{C})$. Then we have the following:

- (1) $e^0 = I$;
- (2) $(e^X)^* = e^{(X^*)}$;
- (3) If $XY = YX$, then $e^{X+Y} = e^X e^Y = e^Y e^X$;
- (4) e^X is invertible and $(e^X)^{-1} = e^{-X}$;
- (5) $e^{(\alpha+\beta)X} = e^{\alpha X} e^{\beta X}$ for all $\alpha, \beta \in \mathbb{C}$;
- (6) $\det(e^X) = e^{\text{trace}(X)}$;

(7) The map $t \mapsto e^{tX}$ is a smooth path in $M(n, \mathbb{C})$ and

$$\frac{d}{dt}(e^{tX}) = Xe^{tX} = e^{tX}X$$

In particular, $\left. \frac{d}{dt} \right|_{t=0} e^{tX} = X$.

(8) If $t \mapsto A(t)$ is a smooth path in $GL(n, \mathbb{C})$, then

$$\frac{d}{dt} [\det A(t)] = (\det A(t)) \operatorname{trace}[A^{-1}(t) \cdot A'(t)].$$

Proof. (1) Observe that

$$e^O = \sum_{m=0}^{\infty} \frac{O^m}{m!} = I + O + \frac{O^2}{2!} + \frac{O^3}{3!} + \cdots = I.$$

(2) Observe that

$$\begin{aligned} (e^X)^* &= \left(\sum_{m=0}^{\infty} \frac{X^m}{m!} \right)^* \\ &= \left(\lim_{N \rightarrow \infty} \sum_{m=0}^N \frac{X^m}{m!} \right)^* \\ &= \lim_{N \rightarrow \infty} \left(\sum_{m=0}^N \frac{X^m}{m!} \right)^* \\ &= \lim_{N \rightarrow \infty} \left(\sum_{m=0}^N \left(\frac{X^m}{m!} \right)^* \right) \\ &= \lim_{N \rightarrow \infty} \left(\sum_{m=0}^N \frac{(X^*)^m}{m!} \right) \\ &= \sum_{m=0}^{\infty} \frac{(X^*)^m}{m!} \\ &= e^{(X^*)}. \end{aligned}$$

Note that the limit above can be interchanged because the complex conjugate $*$: $M(n, \mathbb{C}) \rightarrow M(n, \mathbb{C})$ is a continuous function. Also, the second to last equality holds because for any $A, B \in M(n, \mathbb{C})$, we have $(AB)^* = B^*A^*$ implies $(X^m)^* = (X^*)^m$.

(3) It suffices to show that $e^{X+Y} = e^X e^Y$ since then we would have $e^{Y+X} = e^Y e^X$ and so $e^{Y+X} = e^{X+Y} = e^X e^Y$. But the way we show the above is the same way done for showing that $e^{x+y} = e^x e^y$ for $x, y \in \mathbb{R}$.

(4) From part (3), we see that X and $-X$ commute with one another. Thus, we have

$$e^X e^{-X} = e^{X+(-X)} = e^O = I.$$

Hence, e^X is invertible and its inverse is e^{-X} .

(5) Using part (3) again, we see that

$$e^{\alpha X} e^{\beta X} = e^{\alpha X + \beta X} = e^{(\alpha + \beta)X}.$$

(6)

■